Laboratory 1 Exercises

Learning Objectives

By the end of this lab session, you should be able to:

- Import csv datasets into RStudio.
- Create and run an R script in RStudio.
- Set your working directory.
- Produce and interpret a stem-and-leaf plot.
- Create a histogram with specified class intervals and customise its axes.

Preamble

Before attempting these exercises, make sure that you are familiar with:

A. The RStudio interface;

B. How to create and run an R script;

C. How to set your working directory.

Study materials are available on SurreyLearn under *Course Materials -> R/RStudio*. Please read the Installation and Using RStudio pages to familiarise yourself with these three topics.

For both tasks, you are encouraged to write and execute your commands in an R script rather than entering them directly into the Console. Save your script so that you can revisit your work later.

Task 1. Create a stem plot

Open the *Lab_1_ME.csv* file with the data on military expenditures. Take a moment to look at and familiarise with your data. There are 42 values of military expenditure as a share of GDP are provided for a sample of countries in 2014 (source: World Bank).

Functions you will need

read.csv()	Import a csv file into RStudio.

stem()	Create a stem-and-leaf plot.

Instructions

1. Import the Lab_1_ME.csv file into RStudio using read.csv().
2. Examine the dataset to identify the variable containing military expenditure as a percentage of GDP.
3. Use stem() to produce a stem-and-leaf plot for this variable.
4. Examine the resulting plot and consider what it tells you about the distribution of military expenditure across countries.

Task 2. Draw a histogram

Open the *Lab_1_IQ.csv* file with IQ Scores. Take a moment to look at and familiarise with your data. There are 60 observations. Create 8 bins between 75 and 155 with the width 10.

Functions you will need

read.csv()	Import a csv file into RStudio.
seq()	Generate a sequence of numbers to define the bin boundaries.
hist()	Create a histogram.
axis()	Add or customise the tick marks and labels on a plot's axes.

Instructions

1. Import Lab_1_IQ.csv into RStudio using read.csv().
2. Examine the dataset and identify the variable containing IQ scores.
3. Use seq() to generate the bin boundaries from 75 to 155, with increments of 10.
4. Use hist() to create a histogram with these bin boundaries.
5. Use axis() to display the bin boundaries along the horizontal axis.
6. Add an appropriate title and axis labels to your histogram.